

Excerpts from Special Study Report: LMOC Scalability for Landsat Next Performed Summer, 2023

Notes for LNext MOC:

This document is provided for historical LMOC reference and informational purposes ONLY. This document is ONLY to be used for the purpose of the LNext MOC Studies and IS NOT to be shared or used for any other purposes. It SHOULD NOT be construed as direction or guidance for the LNext MOC. This study was one of many similar activities performed in 2023 during Phase A for Landsat Next, which included a MOC RFI and other special studies on key Ground needs for LNext. The analysis and information included below were based on the state of the MOC in summer of 2023 at the time the study was performed and has not been updated or modified to reflect any changes made since then.

To reiterate, DO NOT utilize this document as an expected architecture for LNext, but only as historical reference of L8 and L9 systems and processes in 2023. The USGS will consolidate the results of all Phase A activities and vendor studies and from them will formulate the LNext MOC.

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1 INTRODUCTION

1.1 Introduction

This document is provided for historical LMOC reference and informational purposes **ONLY**. It **SHOULD NOT** be construed as direction or guidance for the LNext MOC. This study was one of many similar activities performed in 2023 during Phase A for Landsat Next, which included a MOC RFI and other special studies on key Ground needs for LNext. The analysis and information included below were based on the state of the MOC in summer of 2023 at the time the study was performed and has not been updated or modified to reflect any changes made since then.

To reiterate, **DO NOT** utilize this document as an expected architecture for LNext, but only as historical reference of L8 and L9 systems and processes in 2023. The USGS will consolidate the results of all Phase A activities and vendor studies and from them will formulate the LNext MOC.

The Landsat Multi-satellite Operations Center (LMOC) system provides all telemetry, command, trending, flight dynamics, and mission planning services for Landsat 8 and 9 operations, as well as infrastructure, automation, and alerting to support those services. The LMOC has a loosely coupled, highly scalable architecture that was designed to be multi-mission capable and can accommodate new Landsat observatories. Each new mission added to the LMOC will require some development checkout of mission specific capabilities and integration into the multi-mission environment. This special study will discuss the areas of the LMOC where observatory-specific software and processes exist, and what development would be required to support new observatories or operations concepts.

1.2 Proposed Near Term Actions

During the analysis and evaluations a few items have been identified which can be applied now to both improve L8 and L9 operations and prepare the LMOC the for LNext. These projects were evaluated based on the immediate positive impact they can make for the operational costs today.

1.2.1 STK Scheduler Replacement

The software which currently handles contact scheduling, STK Scheduler, still requires manual interactions and verification due to reliability concerns. In the legacy MOC, the FOT developed a Perl script that handled all the scheduling tasks, using the still available view period products for ground contacts generated by the FDS software Meridian. There are open discrepancy uncovered reports (DUR's) regarding the orbit number handling and spurious dropped contacts in STK Scheduler. In addition, some of the operational processes must be handled manually due to reliability concerns, particularly any contact modifications and deletions are handled manually by mission planners. Developing the new tool provides an opportunity to streamline this process, eliminate the discrepancy impacts, have less proprietary code, and save on license fees with immediate effect.

Secondly, one of the new features which is needed for LNext, independent high rate / low rate scheduling, has immediate value for L8 and L9 as well. During MDM anomalies and periods of high image requests, having the ability to scheduling Xband contacts with our ground assets

without having to deconflict the Sband command and telemetry would add functionality and flexibility to the operations teams.

1.2.2 Cloud Migration

There are two main cost drivers with the current LMOC on-premises implementation which we would like to rectify with cloud migration. If these two items are not addressed with LNext, then onsite hardware and software costs will continue to increase to meet the additional processing and storage needs of the LNext Fleet. Figure 1 illustrates the current LMOC cluster utilization. As illustrated the LMOC is oversized for current operations. The CPU, Memory, and Network were all sized to support the strains of Launch and Commissioning which drove upfront equipment costs and continues to drive maintenance costs. The Storage is purchased in discreet 5-year increments and is sized to support the existing data archives and 5 additional years of collection.

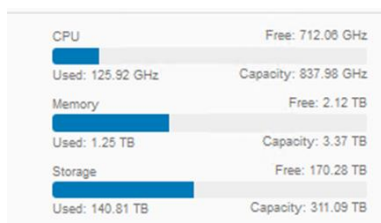


Figure 1: Current LMOC Hardware Resource Utilization

The first cost driver is the compute resources and network / infrastructure to support them. The LMOC needs to export as many offline subsystems, underutilized VM's, and launch surge systems to the cloud as possible. This applies for LNext but also for existing Landsat 8 and 9 operations since most of these subsystems are fleet capable and will be augmented with LNext capabilities. The correct approach for performing this migration without incurring higher cloud service costs can be found in Section 3.3. This will manifest as cost savings in two ways. The first being the overall reduction in hardware needs for the LMOC cluster. As seen in the figures below (Figure 2) the costs for the compute resources and subsequent networking to support it are substantial. Currently the LMOC onsite compute resources could be reduced by half with no cloud migration and with additional cloud migrations possibly $\frac{1}{4}$ of the original. However, if LNext does not support this cloud migration, then the compute needs to support LNext launch within the LMOC will drive these hardware and software costs past the original implementation.

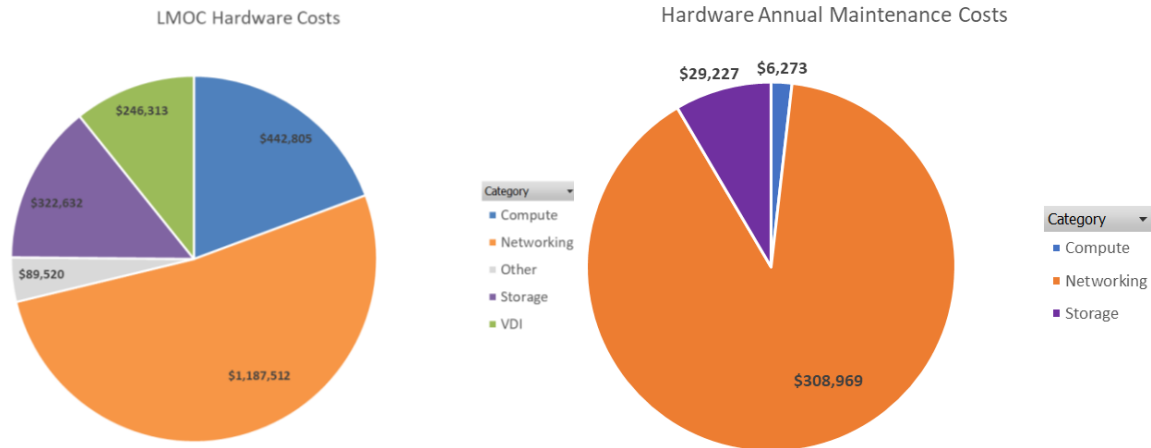


Figure 2: LMO Hardware Costs and Annual Maintenance Costs

Migration of compute resources will also reduce the LMO infrastructure and networking needs. As seen in the figures below the support infrastructure for the onsite systems consumes approximately 44% of the compute resources. Similarly, to how the hardware was sized to support launch, the infrastructure and network was sized to support the subsystems and the surge of users at launch. As we reduce the onsite footprint and export those infrastructure functions to cloud provider services, the onsite infrastructure requirements and needs will reduce.

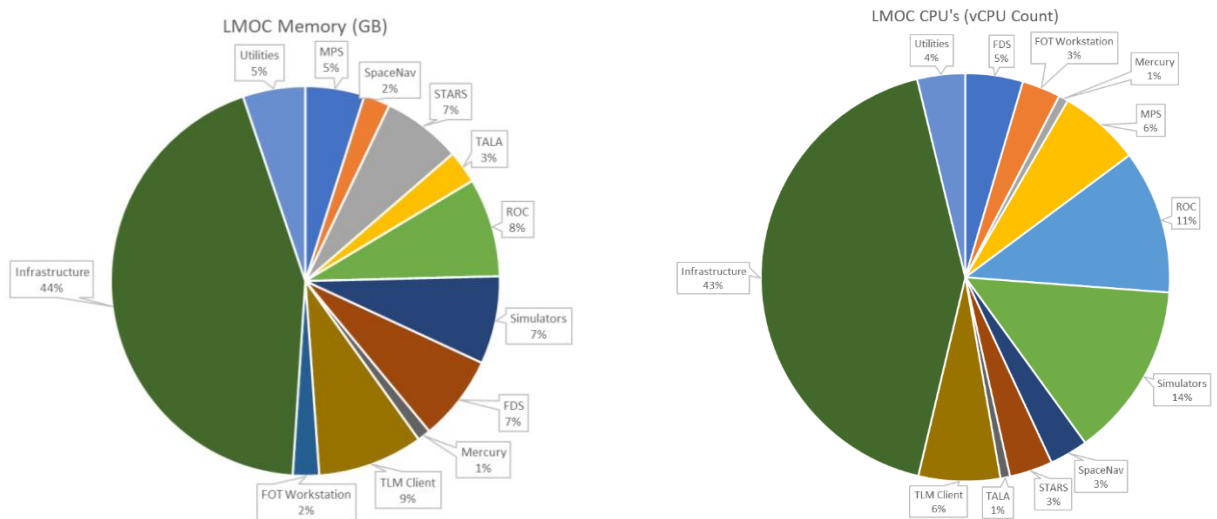


Figure 3: LMO Memory and CPUs

The second cost saving measure with cloud migration is to reduce the onsite data storage at the LMO. As seen in the figure below, over half of the onsite storage array is consumed just with Virtual Machine Backups. This represents almost 150TB of storage and represents a significant cost investment. More efficient and cost-effective cloud solutions are available, and they provide far more data resiliency than an on-prem implementation could achieve. The other 42% of the onsite storage belongs to the STARS trending archive. Currently this archive represents close to

120TB of onsite storage. Within the next 5 years this is expected to exceed 200TB and which will drive backup needs past 250TB. With a few years of LNext data for 3 spacecraft the LMOC could easily exceed 1PB of onsite housekeeping and backup data is infrequently or outright never accessed. This makes it a prime candidate for cloud migration since most data storage costs are for egress, not ingress. More details on STARS cloud migration can be found in section 2.3.12.

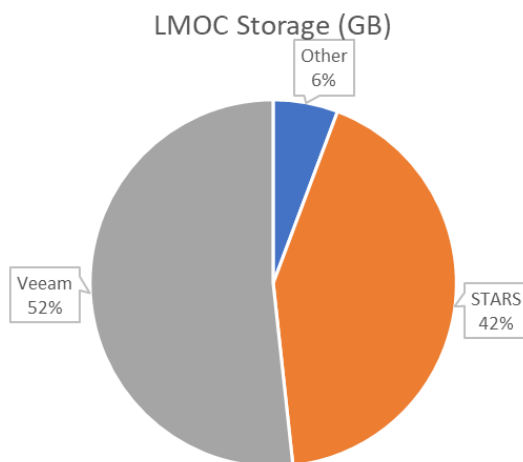


Figure 4: LMOC Storage

1.2.3 Standardizing MPS activity schedule to WRS cycle

A major factor that drives the complexity of mission planning in today's concept of operations is the lack of standardization of activities and how they interact with each other. The result is that each load build is a totally custom build, where we often experience new edge cases due to activities occurring near each other for the first time. To mitigate this, there should be all effort put into scheduling as many activities as possible according to the WRS grid, and therefore increase repeatability and decrease the discovery of new edge cases.

Table 1 gives a summary table of all the L8 activities and how they are scheduled.

Activity	WRS2 Grid Based	Cadence
OLI Dark Calibrations	No, Currently scheduled by CPAW around normal imaging	1-2 per orbit, 2 nd one skipped on orbits with night imaging
OLI Working Stim Lamp	Yes, South Pacific, range of paths, same row each time	Daily
OLI Backup Stim Lamp	Yes, South Pacific, range of paths, same row each time	One per WRS cycle

OLI Pristine Stim Lamp	Yes, Fixed Path/Row	Biannual
OLI Solar Working Cal	Yes, Fixed Path/Row	Twice per WRS cycle
OLI Nonroutine solar cals (Linearity, Pristine, Extended)	Yes, Fixed Path/Row	Replace normal solar working at set cadences. Each of 3 types of non-routine solars occur biannually.
OLI/TIRS Lunar Cal	No	Once per lunar cycle, ~28 days
OLI/TIRS Lunar Cal + Temp Sweep	No	Replaces regular lunar cal once every 3 lunar cycles
Other Non-routine TIRS Cals: FPA Transmit All, ITS Science Mode, BB Stability, 36 min DS, 36 min BB, TIRS Recovery	Yes, Range of Paths, Fixed Row	Variable, driven by lunar cal placement, 3 cals are placed as evenly as possible in between lunar calibrations
IMU Diagnostic	No	Every 14 days
Routine Off nadir collects	Yes, Fixed Path/Row	Salton Sea and Lake Tahoe each once per WRS cycle
Side Slither Collects	Yes, Fixed Path/Row	3 slithers throughout year each with its own target
Orbit Adjustments	No	As needed, typically one DMU per 3-4 weeks, one INC per year

Table 1: Landsat 8 Activity Table

Two items stand out from this list: the non-routine cals driven by Lunar cal placement and the IMU diagnostic telemetry downlinks. The latter is easily handled by moving it to a once per WRS cycle cadence. The former is a bit trickier to resolve and requires buy in from the CalVal team. Obviously lunar calibrations are always going to be independently scheduled without any ties to the WRS grid based on lunar phase angle requirements. However, the other TIRS calibrations are scheduled by the following rules:

1. Place lunar calibrations on the full moon day (or the following day) to meet the lunar phase angle constraint of +5 to +9 degrees

2. Place the 1.5 Orbit Stability, aka Blackbody Stability calibration on the precise day that contains its desired singular WRS path
3. Place the following TIRS cals in between lunar calibrations evenly spaced: ITS Science Mode, FPA Transmit All, 36 min DB, 36 min BB. These calibrations are placed on particular months throughout the year
4. Place TIRS recovery cals whenever there is a gap of more than 8 days between TIRS cals

As the calibration strategy for Landsat Next is developed, the activities should be set to the WRS grid to the highest degree possible. This will simplify the activity deconfliction process of MPS and allow the operations to be maximally automated.

2 LMOC SCALABILITY

2.1 High-Level Architecture

The Landsat Mission Operations Center (LMOC) is the segment of the Landsat Program that performs observatory operations. The LMOC is a combination of 4 separate facilities and includes the entire hardware and software infrastructure needed to support Landsat 8 and 9 Operations. The Primary LMOC is located at Goddard Space Flight Center, which includes both an operations area (LMOC) and the Landsat Mission Equipment Room (LMER) for server and rack housing. During Launch, Early Operations, and Commissioning (LEO&C) a Launch Support Room (LSR) was present to handle Landsat 9 launch surge. This facility was decommissioned shortly after Post Launch Acceptance Review. Finally, the single string Backup LMOC (bLMOC) facility provides physical redundancy in the event of a LMOC multi-thread failure. The bLMOC is located at the EROS Data Center in Sioux Falls, SD with a remote access point at NASA Goddard.

The LMOC consists of six primary services or functionalities:

- Telemetry, Command and Control (T&C)
- Mission Data Management (MDM)
- Mission Planning and Scheduling (MPS)
- Trending & Analysis (T&A)
- Flight Dynamics (FDS)
- Data Management (DMS)

Additional LMOC functionality supports the primary functions, including:

- Ground System Infrastructure (GSI)
- Alerting and Notification System (ANS)
- Government-furnished ground station & spacecraft simulators
- Analytical Modelling (AM)
- Work Management System (WMS): Jira, Confluence, Bitbucket

A high-level architecture of the LMOC with associated data flows can be seen in Figure 5: LMOC Architecture Diagram below.

Looking ahead to the incorporation of Landsat Next, the T&C and MPS system are listed generically as we execute trade studies for software solutions. The ground element is also genericized as there are TBD's for the assets to be used, such as GNE, Near Space Network (NSN). Galaxy, CPAW, and STK Scheduler are shown retained for use on Landsat 8 and 9, and Freeflyer, Mercury, and Stars are shown as retained for the entire fleet.

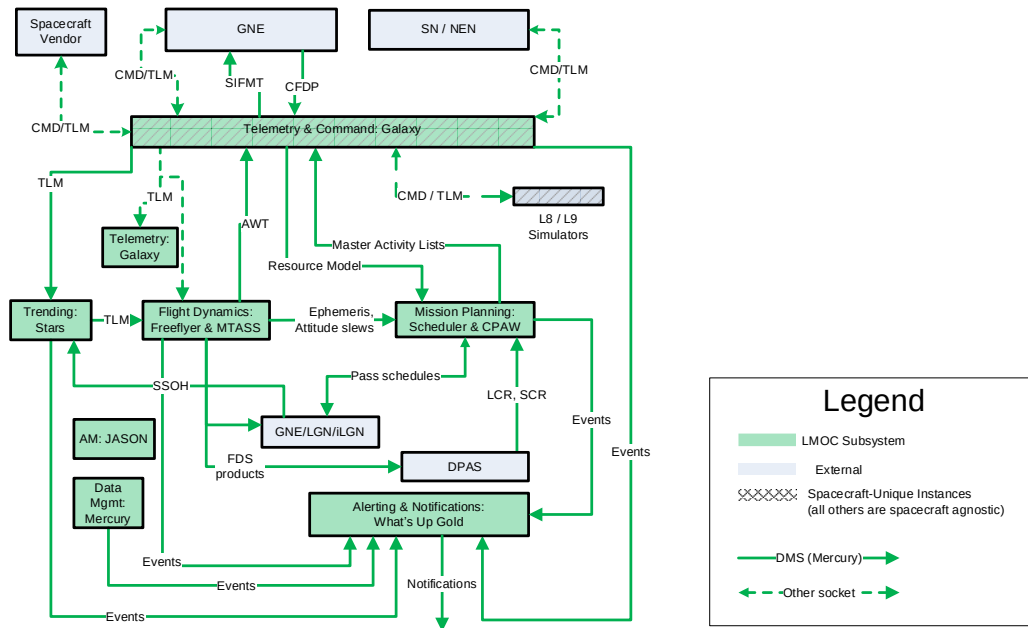


Figure 5: LMOC Architecture Diagram

As a part of this study we will begin to explore the addition of new subsystems to the current LMOC architecture as well as migration of subsystems into the cloud. The LMOC was built on the principle of modularity with loosely coupled interfaces. This design is well suited for adding additional subsystems that interface with existing capabilities and for migration of subsystems to the cloud by extension of our network file systems. As noted earlier, the T&C subsystem and the MPS subsystem currently have the lowest reusability, so a notional architecture has been provided in Figure 6 to illustrate what a modified LMOC infrastructure may look like if new LNext software was brought in for these two subsystems.

These new subsystems can be integrated within the LMOC with little to no disturbance to the other existing systems. Some mercury modifications would be needed to handle the new product flows but these are all done in configuration space and can be easily handled. The FOT still make regular mercury product updates when they have new products or would like to add new recipients to existing products. We would perform the necessary development work to make the new subsystems compatible with the existing FDS and DPAS products to concentrate development efforts on the new systems instead of opening source code on existing proven

systems. Since all of the LMOC's interfaces are file based, the integration of new subsystems can be handled in a straightforward fashion.

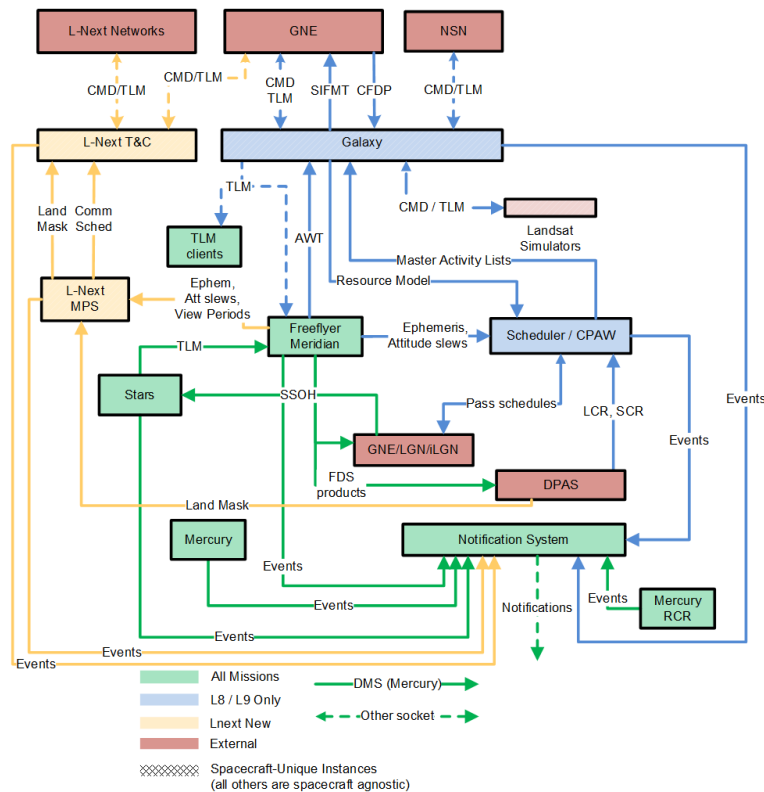


Figure 6: Notional L8/L9/LNext : Updated LMOC Architecture Diagram with LNext flows

A medium resolution set of flow diagrams describing the full end to end process for the generation, approval, and activation of the current command loads are given below in Figure 7 through Figure 12.

Nominal Planning Flow Load Planning

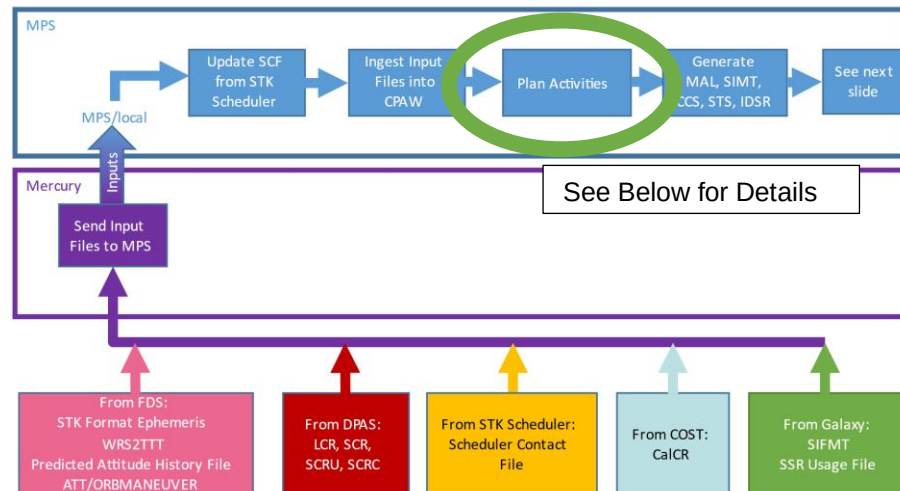


Figure 7: Load Planning Flow Diagram

Nominal Planning Flow Load Building

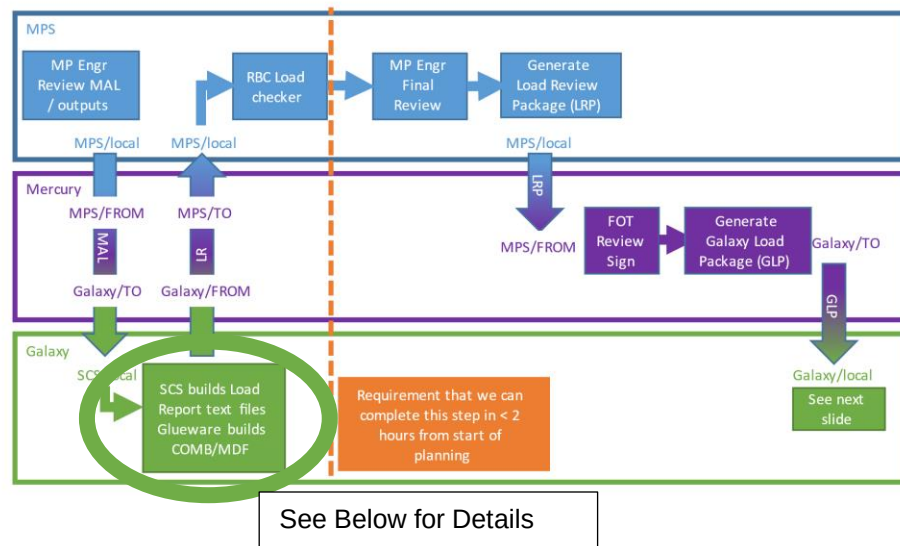


Figure 8: Load Building Flow Diagram

Nominal Planning Flow Load Execution

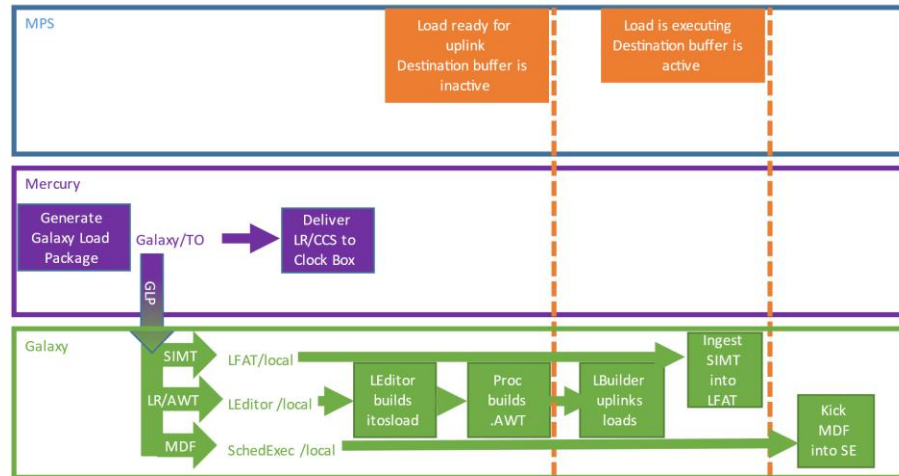


Figure 9: Load Execution Flow Diagram

Nominal Planning Flow Activity Planning / CPAW Deep Dive Pt 1

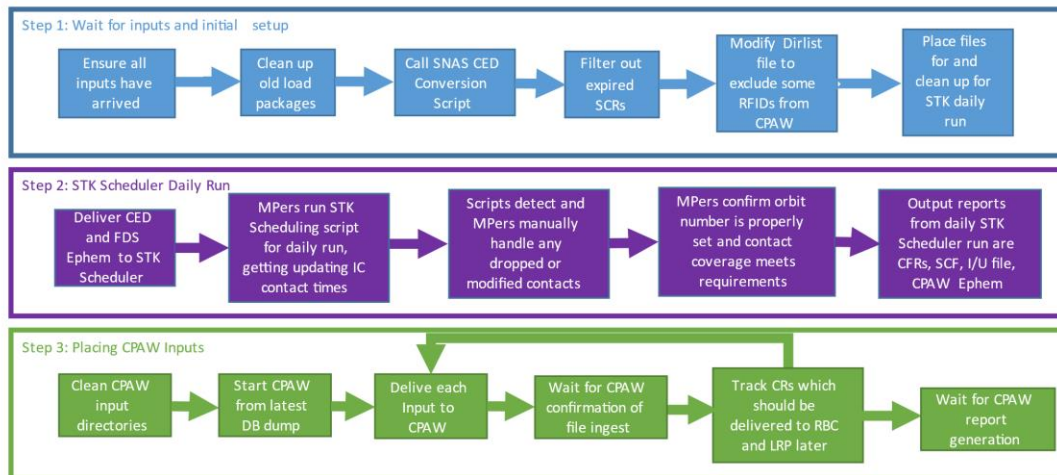


Figure 10: CPAW Load Building Details Pt 1

Nominal Planning Flow Activity Planning / CPAW Deep Dive Pt 2

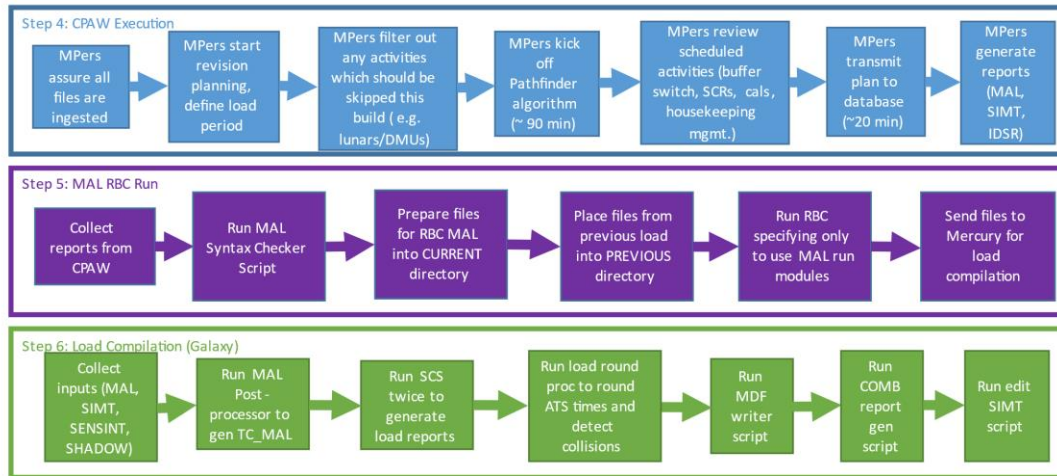


Figure 11: CPAW Load Building Details Pt 2

Nominal Planning Flow Activity Planning / CPAW Deep Dive Pt 3

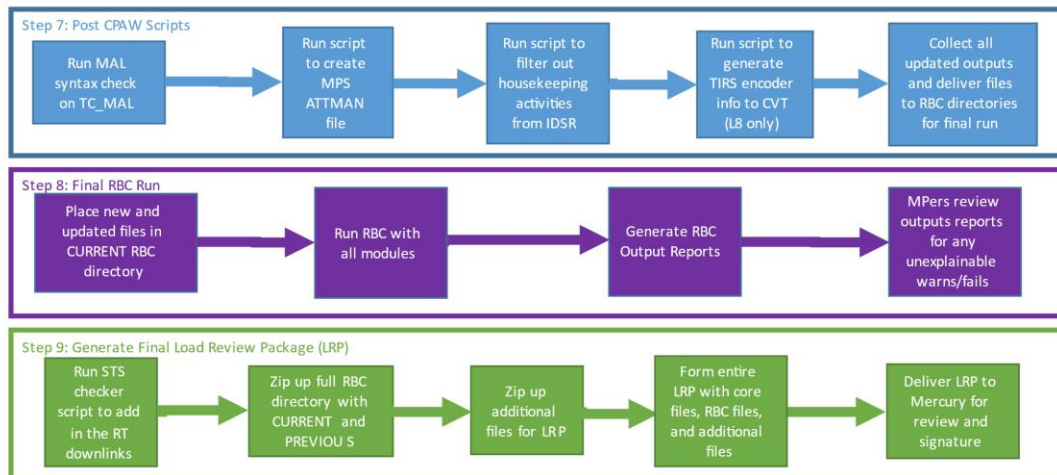


Figure 12: CPAW Load Building Details Pt 3

The majority of the tasks listed in the above figures are automated. Those which are not are noted by the description indicated tasks done by the mission planners (MPers).

2.2 Telemetry and Command Subsystem Scalability

The Telemetry and Command functionality is provided by the GALAXY software developed by the Hammers Company. The Hammers Company also develops and maintains the Mission Data

Management tool, commonly called the Landsat File Accounting Tool (LFAT) which is paired with the GALAXY System. GALAXY also contains the Stored Command Compiler (SCS) which compiles and builds the observatory command loads. The T&C system establishes, configures, and maintains the data communications paths between the LMOC components and the Networks (Ground Network Element [GNE], Near Earth Network [NEN] and the Space Network [SN]) or Landsat Simulators. The Telemetry function processes Housekeeping (S-band) real-time telemetry to assess spacecraft state of health and supports command functions. This function evaluates any housekeeping (HK) data: real-time (R/T), stored state of health (SSOH), and station playback (P/B). The Command function generates, manages transmission, and verifies all Landsat commands. This function supports command encryption, authority management, and constraint checking prior to transmission of the commands to all Landsat observatories. Command verification is performed following issuance of commands. Command loads are built to implement observatory activity schedules and flight software updates.

2.2.1 Galaxy Pressure Points

The following sections will look at each functional area within the Telemetry and Command system with an explanation of the readiness assessment for each. This section will give an overview of the major pressure points that impact both current L8/L9 operations as well as those which need to be addressed to support the LNext triplet. The augmentation plan for the LMOC will include:

1. Command Encryption: We have experienced six GEDU crashes since 2023 while operating Landsat 9, which all resulted in missed ground contacts and emergency anomaly response from the realtime ops and GSE teams. This is related to the hybrid infrastructure of a virtual machine ROC interfacing with a physical EDT card.
2. Mission Data Management: The LNext conops will require a total rework of the LFAT tool..
3. Non-automated tasks: In an effort to reduce staffing the operations should be automated to the highest degree. There are a few tasks which has remained manual operations in the LMOC due to an absolute minimum risk stance or due to lack of return on investment to automation. These include:
 - a. Orbit adjustments: These could be automated today, there's nothing technical in the way. The reason it's not currently is minimum risk posture. The savings in terms of team effort is high between the simulations and realtime support during the burn.
 - b. Monthly EOP (drag and timing parameters) uplinks: could be automated, trigger off the delivery of the EOP file to the ROC. Automating this would save a few hours a month so the return on investment is low.
 - c. Proficiency exercises (connectivity tests with stations, handover contacts, blind acquisitions, low rate passes): To some extent these activities cannot be fully

automated. Handover contacts have to be manually coordinated with the ground station operators. But the team is implementing processes to ease the planning aspects of these and can continue to refine these processes going forward.

- d. Anomaly responses: Again this cannot be fully automated across the board, but with an increase in the use of configuration monitors and telemetry monitors some anomaly responses can be automated. This has already been used to effect for a few anomalies that have occurred after launch of L8 and L9, including PIB-T lockup mitigation for L8, RFI response for L9, and SSR bad block response for L9, among many others.
4. Database Translation: Assuming the T&C database input file format will be significantly different than the format used by Northrup Grumman for L8 and L9, then there will be significant rework on the PRDS translation tool.
5. Flight Procs: With the anticipated little heritage with Landsat 8 and 9 for spacecraft flight hardware, almost no reuse of flight procs for Landsat Next.

2.2.2 Fleet Operations Considerations

As the LMOC expands to support a growing fleet there are a few major considerations as it pertains to the T&C system design and development. First is the consideration of the context structure for the fleet. There are currently three levels that are considered for the context (see Figure 13 below). All the operations products (procs, pages, sequential prints, configuration monitors, record language files, other config files) are assigned a context (L08, L09, FLEET, L8L9, ALL) which is determined by its directory location within the Galaxy repository.

Whenever the Galaxy application is started, a context is specified. Only the lowest level green boxes are permitted as selectable contexts in which to run Galaxy (L08, L09, or fleet). For each of these selections, the Galaxy session has access to operations products which are specific to that context, but also inherits the operations products for any linked context at higher levels. For example, L08 context has access to L08 products, L8L9 products, and ALL products.

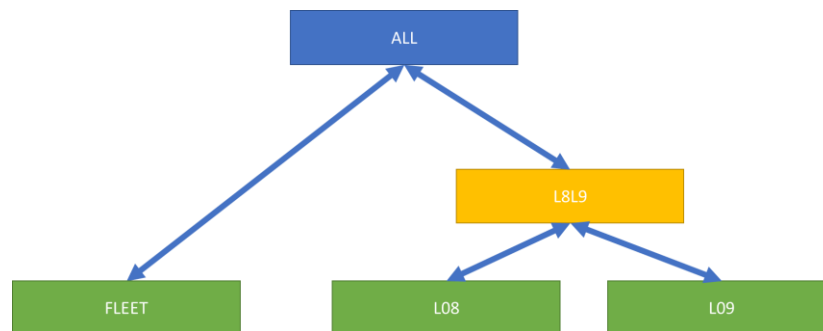


Figure 13: Current LMOC Galaxy Context Architecture

The bottom right corner of this chart shows L8 and L9 individual products and shared products in the L8L9 bin. The FLEET context is used very rarely, but the intent was to provide a context which could connect to telemetry streams for either L8 or L9. It has no access to any commanding procs, but has a few pages which contain both L8 and L9 telemetry. It primarily is configured for displaying ground configuration information. The ALL context has no Landsat specific operations products and is the location for Galaxy application-level products (start up and tear down scripts for example).

The proposed context structure for the combined L8, L9, LNext fleet is shown in Figure 14. Notice that the second yellow layer differentiates the two generations of the fleet. There are likely to be a significant amount of operations products which live at this level. Most of the current L8L9 products are unlikely to be applicable to the LNext fleet, and will remain in the same bin. And most of the LNext flight and ground procs are likely to be consistent for the entire LNext triplet and live in the LNext bin. The LNext1,2,3 boxes are expected to contain only a small number of deltas for each spacecraft within the triplet (e.g., differences in telemetry limits).

The LSAT level is added in the gray box to contain products relevant for all five spacecraft. The FLEET context remains for telemetry monitoring of multiple spacecraft. The ALL context also remains for Galaxy application-level products.

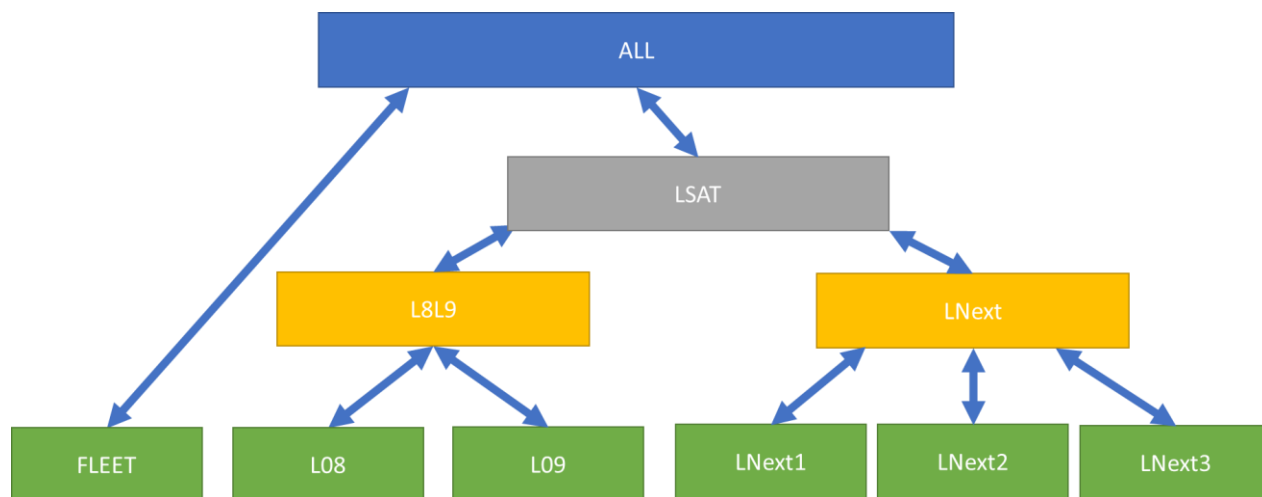


Figure 14: Proposed Galaxy Context Architecture for Augmented LMOC

Another fleet operations consideration is how we manage the fleet mapping to real-time operations consoles (ROC's). Currently the LMOC has enforced that each ROC is able to command only one spacecraft, L8 or L9, with no crossover potential (See Table 2).

ROC	Commanding Ability
LMOC ROC A	L08 (Primary)
LMOC ROC B	L08 (Secondary)

LMOC ROC C	L09 (Primary)
LMOC ROC D	L09 (Secondary)
LMOC ROC E	L09 (Contingency)
LMOC ROC F	L08 (Contingency)
BLMOC ROC A	L08 (Tertiary)
BLMOC ROC B	L09 (Tertiary)

Table 2: ROC-Spacecraft Mapping in Current LMOC

2.2.3 T&C Database

The T&C database functionality involves the ability of the ground system to translate a command and telemetry database from the spacecraft and instrument vendors into a format useable by the T&C system. For Galaxy this means generating record language files (rec files) for the telemetry mnemonics, command mnemonics, telemetry limits, and mnemonic conversions.

For L8 and L9, the format of the T&C database files was in XML coding, one file per command and telemetry mnemonic. The XML file would prescribe all the pertinent information for that mnemonic: APID, byte location, variable type, conversion polynomial if applicable, etc.

One further method to improve efficiency is to have a single set of database files for the LNext triplet which then only needs to be translated once. If there are any small changes between each of the triplets, those can be captured in separate rec files which are then compiled into a Galaxy database specific for that spacecraft.

2.2.4 Commanding Script and Procs

The effort of writing and/or updating the procedure (Procs) and commanding scripts for Landsat 9 launch was a significant percent of the overall Galaxy I&T effort. LNext has significantly less heritage to the on-orbit observatories than L9 did to L8. Therefore, it is largely expected that the majority of the flight procs will have to be rewritten. However, the team can leverage the established process for formatting the procs, testing them, and managing configuration.

In addition to flight procs, there are ground procs which are largely multi-mission in nature. These procs would require significantly less rework than the flight procs. Overall, L8 has 465 flight procs, while L9 has 439. For ground procs there are 45 multi-mission procs, along with 24 specifically for L8 and 19 for L9.

In preparation for the L9 launch, 421 procs were written and certified for operations as the team successfully executed 8 MRTs, 7 GRTs, and 5 Mission Simulations in a span of 10 months.

2.2.5 Telemetry Archive

Galaxy's telemetry archive subcomponent contains no mission-unique elements and could be readily used with the new LNext triplet of observatories without modification. Figure 15 shows telemetry paths for real-time data, RAM SSOH telemetry downlink via S-band, and HKDB file downlink via X-band, applicable to L8 and L9. Depending on LNext comm systems (especially Ka-band vs X-band), some changes to the inputs to the network layer may be required, but the LMOC systems can easily be adapted for maximum reusability and cross-mission support.

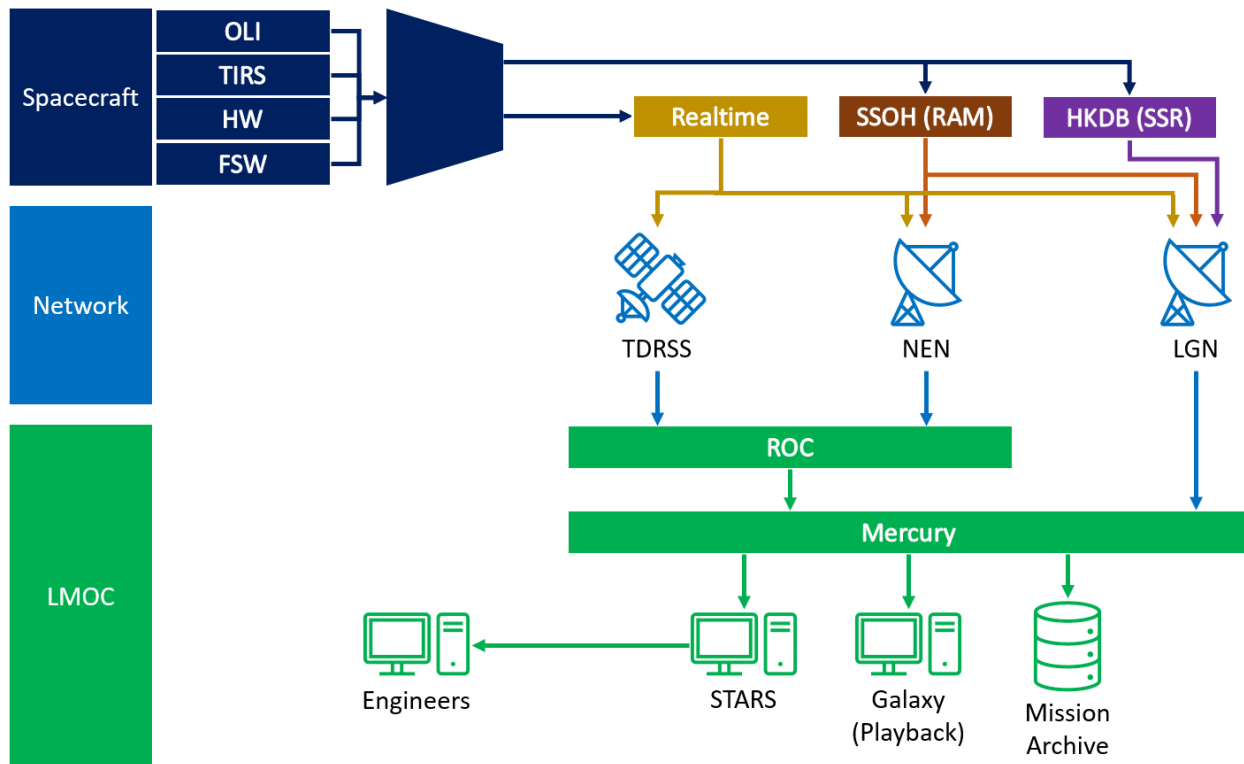


Figure 15: Realtime, RAM SSOH, and SSR HKDB Telemetry Paths, Use, and Storage

2.2.6 Telemetry Decom / Display

Galaxy's decom and telemetry display subcomponents are driven by the T&C database and contain no mission-unique elements which make it ready for the LNext observatories without modification.

2.2.7 Command Generation

Galaxy's command generation subcomponent is driven by the contents of the T&C database and contains no mission-unique elements which make it ready for the LNext observatories without modification. A high-level command flow diagram is shown in Figure 16, with functions in dark blue provided by Galaxy in the LMOC (and available in most other T&C software systems). One important note is that Galaxy does not perform command encryption, but it does encapsulate encrypted command data for network routing; it is the sole input and output for the ground encryptors.

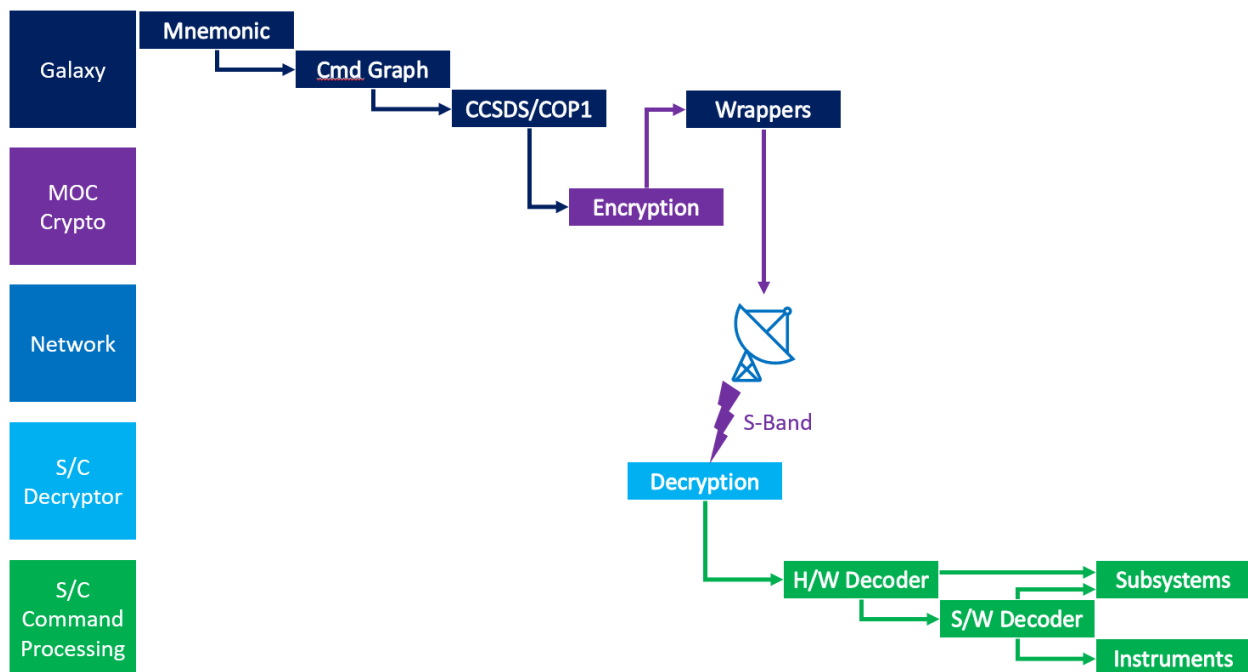


Figure 16: Command Generation, Transmission, and Processing

2.2.8 Command Encryption

The LMOC utilizes hardware command encryption solutions for both Landsat 8 and 9. These hardware encryptors are interfaced to the Realtime Operations Controller (ROC) virtual machine.

Another factor to consider during the augmentation plan of the LNext is to ensure there are enough encryption keys to support the full expected lifetime of the fleet under the key management plan.

2.2.9 Limit Monitoring

Galaxy's limit monitoring subcomponent is driven by the contents of the T&C database and contains no mission-unique elements and could be readily used for the LNext observatories without modification.

2.2.10 Schedule Executor

Galaxy's schedule executor subcomponent is driven by a schedule file generated by the mission planning system and contains no mission-unique elements which make it ready for the LNext observatories without modification.

2.2.11 Mission Data Management

Mission Data Management (MDM) was a significant portion of both the system development and the I&T effort for T&C. The current high-level MDM concept is shown in Figure 17, which shows the sequence of key events and file transfers in the generation of a science product for Landsat 8:

- MPS generates command load inputs and a matching Scene Interval Mapping Table (SIMT), which will correlate SSR file ID numbers with path/row intervals.
- The ROC generates the spacecraft command load and ingests the SIMT file into LFAT (operating on the ROC).
- The spacecraft collects the image and records it to the SSR.
- On a realtime contact, the spacecraft transmits science data over X-band directly to the ground station, while sending the directory listing of files in the recorder to the ROC.
- The ground station sends the list of acknowledged files (ACKs) to the ROC.
- LFAT correlates the lists of recorded files and science intervals, generates the Scene Interval File Mapping Table (SIFMT), and sends it to EROS for use in reconstructing the science image files from the raw X-band files transmitted from the ground station.

This process is much the same for Landsat 9, with a key terminology difference: the Image Window Table (IWT) command load for L8 is called the Mission Absolute Time Sequence (MATS) on L9, though both contain timestamped record commands and other science activity sequences. For the sake of simplicity, a number of internal steps are omitted, to allow the reader to concentrate on interfaces between mission elements.

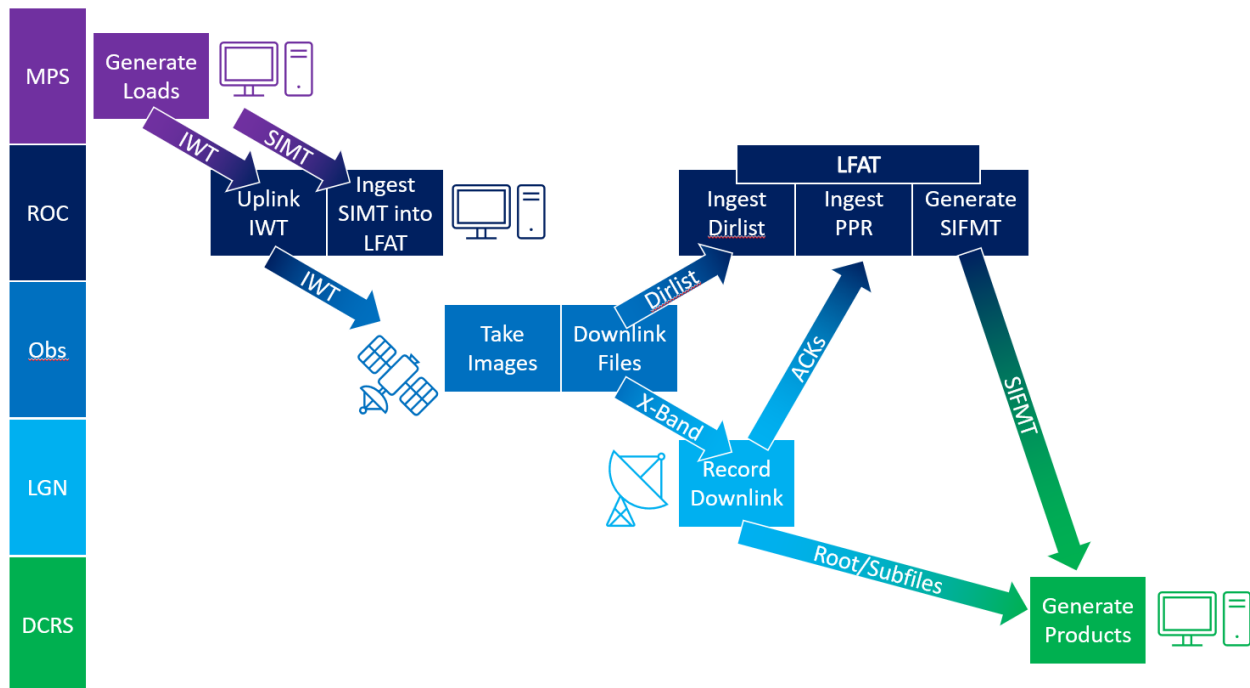


Figure 17: Landsat 8 Science Scheduling, Collection, and Product Generation High-Level Overview

Figure 17 also illustrates the importance and complexity of LFAT, as it provides correlation between three different science data accounting systems:

- The spacecraft SSR, which identifies files only by Root File ID (RFID, the SSR file ID name commanded by the IWT/MATS)
- The ground science collection plan in MPS, which identifies intervals to be collected from various inputs and assigns RFIDs to the record commands in the IWT/MATS
- DCRS, which works on interval IDs and has no direct insight into RFID planning. LFAT is crucial to this science processing by providing the SIFMT that maps RFID from the ground station and scene IDs planned in MPS.

2.2.12 Load Compiler

The reuse for the load compiler software (SCS) is very high, however the development of the activity definitions which link an activity to the corresponding sequence of commands to include in the Absolute Time Sequence (ATS) stored load are very mission dependent. The main difference is that usually the demonstrations of the product to the vendor is done by tabletop reviews of the ATS and simulation logs as opposed to live simulator demonstrations for practical time reasons.

2.3 Trending and Analysis Subsystem Scalability

The Trending & Analysis (T&A) functions process near-term and long-term observatory telemetry data from all Landsat observatories. This function is used to trend and analyze the performance of each observatory for state of health monitoring using all collected housekeeping data (e.g., potential problems with the observatory attitude, power, temperature, or other subsystems). T&A provides the capability to report data in various formats to meet engineering needs. The LMOC already has the ability to access this functionality remotely.

2.3.1 Telemetry Archive

Archival of telemetry data in the STARS database is driven by the mission T&C database (the same database used by Galaxy) and therefore requires no modification to support additional missions. Incorporating LNext into the LMOC would require obtaining more storage arrays, either physical hardware or via cloud storage services. Currently Landsat 8 and 9 are generating approximately 20.5 TB of housekeeping data annually combined. The LMOC storage system is currently sized to accommodate L8/9 data through 2028.

LNext would likely require its own instance of STARS with a separate database from Landsat 8 and 9. This would mean that there would be potential of process collision between the two generations. It also is likely required from a practical standpoint in terms of size of database, failovers, database scrubbing, etc.

2.3.2 User Interface

The STARS user interface contains no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.3 Data Export

The STARS data export features contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.4 Data Ingest

To support simultaneous ingestion of L8 and L9 housekeeping data, the STARS ingest software has been threaded. Theoretically a single STARS server could be resource loaded to handle additional missions. The system has demonstrated up to five parallel ingests during operations (concurrent L9 ops, 3 parallel L8 legacy ingests, L9 simulation). STARS current ingests a rate 14x the housekeeping rate for L8 and 12x for L9 and has been shown to perform well even with months of continual 100% duty cycle usage. To routinely support five spacecraft plus simulations plus readiness testing, STARS would likely need a resource increase in RAM and CPU cores.

However, there is a trade study assessing a single STARS server vs. separate servers for L8/L9 and for LNext. Having separate servers would allow partial failovers of operations from LMOC to BLMOC. In addition, it would enable an upgrade in the Postgres database version to the latest version without potential impacts to historical data. Separate servers mean higher licensing fees. A hybrid approach is also under consideration where LNext begins in a new cloud-based server and then L8/L9 merge into that server (or its own cloud-based server) when the on-prem hardware is due for a refresh.

There are two modes for data ingests and how to keep servers in sync between LMOC, BLMOC, and during LSR (during LEOC). The preferred method is to have a primary and backup system and sync the database from primary to backup. The second method is to set up parallel ingests to all environments, which requires increased monitoring to ensure the servers stay in sync. The default configuration would be that the LMOC is primary and BLMOC is receiving database syncs from the LMOC. During most BLMOC failovers, the configuration would be toggled to parallel ingests in both environments. This would remain until operations is restored in the LMOC. The hybrid approach allows each spacecraft to failover when the comm schedule determines it to be convenient. For longer duration LMOC outages, once all spacecraft were transitioned over, the BLMOC would instead become the primary and would provide syncs back to the LMOC.

2.3.5 Limit Monitoring

The STARS limit monitoring and reporting features are all driven by limits in the T&C database and therefore contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.6 Event Message and Command History

Command history validation is performed using the load report and the command echoes from housekeeping telemetry archives. A simple script is utilized to perform this check. It is anticipated that only small modifications to this script will be necessary to support new missions.

2.3.7 Statistics

The STARS statistical computation and reporting features contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.8 Routine Products

Generation of routine trending products from STARS is not driven by any mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.9 Data Quarantine

The STARS data quarantine features contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.10 Housekeeping Completeness

Currently mission housekeeping data is stored onboard the L8/L9 observatories on the Solid State Recorder (SSR) in file format, and is played back to the ground as a CADU file, which is then captured by the ground station and sent to the LMOC for processing. This requires that the LMOC determine the time period that the data covers from the command load and the packet sequence.

This process is handled by housekeeping completeness software in STARS. While this tool is generic enough to be used on future missions, the process would be more efficient and consistent if instead the onboard CCSDS File Delivery Protocol (CFDP) software modules were used to package and downlink housekeeping data files.

2.3.11 Data Replication

The STARS data replication features contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.3.12 Cloud Migration

There are two approaches for migrating STARS to the Cloud. The first approach involves little to no software modifications but will yield little to no cost savings. The second approach will involve some modest software improvements and enhancements but may yield significant savings with hardware and data storage costs.

The first approach is to stand up a second STARS system in the cloud to be used as a long-term cold archive. This STARS instance would be a lift and shift application and we would anticipate rare access by design. Once every year the cold archive would be powered up to ingest the previous year's data, this ingestion would coincide with the Mission Annual Review when the FOT typically looks at performance metrics across the life of the mission. Once all the data had been ingested and the cold archive was up to date, annual metrics would be generated and the VM would be powered off. The on-prem STARS system would currently be holding 2 full years of mission data and could be pruned down to 1 year of mission data and continue serving FOT

needs. The offline STARS system in the cloud could still be accessed by the FOT, it may just require additional time to bring the system online, which is acceptable for long term trending. The On-Prem system would maintain a lean storage archive improving performance of the more often looked at data.

The second approach to STARS migration to the cloud involves software updates and enhancements to make the application more cloud native. In this implementation the entire STARS trending system and archive would be moved into the cloud and no on-prem system would exist. The STARS software would be containerized, and the STARS data files would be reformatted to cloud native storage formats.

2.3.13 Machine Learning

Telemetry Analysis with Learning Algorithms (TALA) is an anomaly detection system. It was tested on data from the L8 Legacy MOC and implemented in the LMOC for both L8 and L9 missions. TALA uses Python's scikit-learn machine learning packages to evaluate deviations in telemetry based on previous mnemonic behavior. TALA is trained on past data to learn expected behavior, then evaluated on new data to detect anomalies. TALA is designed to detect more subtle behavioral changes and anomalies than we could detect with our other tools (config check, telemetry limits).

TALA is fully fleet capable: each spacecraft has config files defining the predictive and monitored mnemonic lists. Scaling up to support the LNext fleet is simply a matter of defining new mnemonic lists for the new spacecraft and setting new cron jobs to train and evaluate the data.

2.4 Data Management Subsystem Scalability

Product, event, and report handling is the process by which the products, reports and events generated by the LMOC Subsystems are delivered to the intended recipients. Products are considered any file or information set generated by a LMOC subsystem intended for use by another LMOC subsystem or external element. Reports are human readable documents or displays intended for use by the Flight Operations Team (FOT). In this context, events are considered the warning or error messages generated by the various subsystems and used to alert the FOT. The ability to monitor these event messages remotely is a function of the LMOC system.

2.4.1 Controlled Repository

The Mercury Controlled Repository (CR) is organized by mission and the Mercury model. Automated scripts are used to build the repository structure given a mission name and can be used to add additional missions.

The I&T rework needed in this area is focused on establishing new Mercury mounts and network file systems (NFS's) for the LNext fleet, as well as controls to ensure proper segregation of products between the missions.

2.4.2 Product Approvals

The Mercury Product Approval processes contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.4.3 Controlled Repository Management

The Mercury CR Management tools contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.4.4 Web Portal

The Mercury Web Portal is a HTML representation of the archive and contains records on product movements throughout the LMOC. There is no mission specific code in the web portal, and it could be used for the LNext observatories without modification.

2.4.5 Mercury Model

All products within the LMOC are labeled by the Mission and Observatory source (Flight or Simulator). A new suite of products would be generated for additional Landsat missions. These product naming conventions would need to be added to the Mercury model. The Mercury model is a FOT configurable XML file defining all the product movements in the LMOC.

The I&T reuse is driven by updates required to the Mercury model for new products defined by the next generation of the mission. There were some significant new products and interfaces established during the generation of the LMOC and the expectation is that there will be a more modest adjustment for LNext.

All products within the LMOC are labeled by the Mission and Observatory source (Flight or Simulator). For existing products in the Mercury model, the regular expressions must be updated to add in the new spacecraft. The Mercury model is a FOT configurable XML file defining all the product movements in the LMOC.

2.4.6 Product Processing Glueware

There are a number of support scripts that reside within the Mercury subsystem which are used to manage modifications to products to meet ICD requirements or to support operations ease automation. With a modest expectation of new products required for the LNext fleet, some of these glueware scripts may require rework.

2.4.7 Cloud Migration

The Mercury controlled repository can easily be migrated to the cloud since it is all file based. Once products hit their pruning age in Mercury (which is approximately 30 days) they should be moved to cold storage in the cloud such as S3 Glacier, EFS or FSx. These files are accessed extremely infrequently and could be stored in a much more cost-effective manner.

2.5 Mission Planning Subsystem Scalability

The current software solution in operations in the LMOC for mission planning includes the following components:

- Systems Toolkit (STK) Scheduler: Handles contact scheduling and forecasting, including Landsat Ground Network (LGN), Near Earth Network (NEN), Space Network (SN), and International Collaborators (IC). The base software is proprietary provided by Orbit Logic, but the Landsat Interface and scheduling scripts are not proprietary.
- Collection Planning and Analysis Workstation (CPAW): Handles activity deconfliction, image collection planning, routine calibration scheduling, solar/offnadir/side slither attitude maneuver planning, MP report generation. The base software is proprietary provided by Orbit Logic, but the Landsat Plugin and default task orders are not proprietary.
- Calibration Operations Schedule Tool (COST): Non-proprietary FOT-built Python script for calibration scheduling, forecasting, and deconfliction.
- Report Builder and Checker (RBC): Non-proprietary FOT-built package of Python scripts for automated load constraint checking
- Master Activity List (MAL) Post-processor (MPP) and Edit_SIMT scripts: Non-proprietary FOT-built Python scripts for automated load postprocessing
- File Monitor: Non-proprietary FOT built Python package for MPS automation. Handles all file manipulation, triggers some post processing scripts, and generates the final load package for FOT review.

The planning and scheduling tools in the LMOC will require the most changes when adding new missions. Since mission planning is the primary implementation of the flight segment operations concept, changes in imaging and calibration concepts, or changes to the WRS grid will have impacts to the planning and scheduling tools and require re-development. Since it is the most vulnerable to mission changes, it is also the area where it would be most logical to simplify by implementing more mission planning capabilities onboard the observatories. By migrating some key functionality and processes onboard we can remove some of the mission-unique aspects from the LMOC MPS tools and make them more readily scalable for additional missions.

The following table provides a high-level overview of the key functionality provided by CPAW & STK Scheduler, its potential reuse for additional missions, and some particulars for where mission specific functionality has been configured or developed.

Functionality	Particulars
Contact Scheduling	A single observatory would have high reuse A constellation would potentially require introduction of internal resource deconfliction assuming we were splitting pass windows
Collection Planning (Calibrations)	Calibration OpsCon and planning has been highly tuned to OLI/TIRS, new architectures would require software re-work
Interval Planning (NADIR Imaging)	Given the assumed orbit changes, we expect a new WRS grid or something different The WRS-2 Grid has been developed into the CPAW architecture so new designs would result in software development
Collection Requests	Natively multi-mission, little to no change required
Housekeeping Operations	Re-development will be required to support the design changes made to the observatories, unless housekeeping operations are similar or identical to L8/L9
Recorder Management Planning	Re-development will be required to support the design changes made to the observatories, unless housekeeping operations are similar or identical to L8/L9
Activity Compilation	Natively multi-mission, little to no change required
Command Load Verification	New observatory constraints would need to be written into the
Mission Planning to DPAS	Natively multi-mission, little to no change required
Calibration Opportunity Scheduling Tool	New Calibrations will need to be incorporated into the COST tool.
Attitude Planning	Multi-mission design, unless attitude profile and maneuvers are drastically different from L8/L9, development should be minimal
MAL Postprocessing	Many modules specific to L8 or L9 but also will be mostly OBE for Lnext

Table 3: MPS Scaling and Reuse by Function

2.5.1 Contact Scheduling

Given a triplet of observatories for LNext, it is anticipated that more LMOC internal station management and deconfliction activities will be required. The lower altitude and longer repeat cycle for the LNext triplet compared to L8/L9 dictates that there will be periodic conflicts for shared resources due to the relative precession of the two generations of observatories. With the assumption that some of the ground assets will be shared across the entire Landsat fleet, the LMOC MPS tools may need to be able to segment up station contacts and perform internal mission prioritization. This is not a functionality currently available in the LMOC.

The switch to Ka-band antenna (from X-band) for the high-rate downlinks may result in a significantly different set of ground antenna resources supporting LNext, which would somewhat relieve the impacts and frequency of the shared resource deconfliction. However, the RFI material proposed 5-8 S-band contacts and 10-12 Ka-band contacts per day per LNext bird. Separating the S-band assets from the high-rate downlink assets (X-band / Ka-band) would be new functionality for STK Scheduler.

The primary method to handle this would be to define separate S and Ka band resources for ground stations capable of both and to schedule them independently per spacecraft. To support

this conops, there would be development work on CPAW when reading in the Scheduler Contact File.

2.5.2 Routine Calibration Scheduling

Planning of the routine instrument calibrations is currently performed by the Landsat Model in CPAW. For the on-orbit fleet, this includes Operational Land Imager (OLI) shutter collects (1-2 per orbit) and Thermal Infrared Sensor (TIRS) deep space and blackbody collects (2 per orbit for Landsat 9 only). All other calibrations are scheduled by COST. Since the routine calibrations are specific to the OLI/TIRS instrument suites, it is unlikely that the software can be used as-is.

2.5.3 Interval Planning (NADIR Imaging)

CPAW still must read in the Long-term Data Acquisition Plan (LDAP) Collection Request (LCR) in order to deconflict LCR imaging against calibrations from COST and Special Collection Requests (SCRs) which are still commanded via Absolute Time Sequence (ATS) loads. MPS still must track the LCR region and when each observatory enters and exits to properly determine the commanding sequence for the SCRs. This hybrid LCR/SCR approach leads to dramatically reduced load build durations and a simplified stored command sequence overall.

2.5.4 Collection Requests

The Collection Request interface with DPAS could be re-used for future missions. If the WRS-2 Grid is no longer in use, there may be some User Request file format changes, but these are relatively simple and interface changes could be made.

2.5.5 Housekeeping Operations

Current housekeeping operations onboard Landsat 8/9 include S/X-band Transponder management, SSR management, Housekeeping Data Recording, Load Buffer Swapping, Orbit Station keeping, and other miscellaneous operations as needed. These activities are specific for the LEOStar-3 spacecraft bus and would most likely require re-writes to reflect the new mission(s) logic. The interfaces for this software however are fairly mission agnostic and could be reused as is.

2.5.6 Recorder Management Planning

Recorder management monitors the utilization of the SSR recorder and ensures that sufficient downlinks are present to avoid exceeding SSR thresholds. This model has been tuned to Landsat 8 / 9 operations and would require some configuration to accurately model the new instrument(s), data downlink rates, and new stations that would be added for the new missions.

2.5.7 Activity Compilation

The process of compiling all observatory activities into a single master activity list is not mission specific, however the process to schedule, compile, and deconflict them does take several hours. The Software tasks will take up to 2 hours for complex loads and the manual cross checks by the FOT may take up to an additional hour. Currently there is sufficient margin in the day to run Landsat 8 and 9 sequentially on a single instance of CPAW without causing any downstream timing conflicts.

With the LNext conops however, the compilation of activities is made much more straightforward. Managing SCRs and configuring the FSW to automate their scheduling appears to be more of a realtime ops automation task than a MPS task. The only activities that have to be built into the ground ATS are non-routine calibrations, orbit adjustment burns, and other special activities.

2.5.8 Command Load Verification

The scaling effort for command load verification covers any new constraints that CPAW needs to verify before generating the MAL. It also covers the I&T work to update RBC to verify all stored load constraints and opscon checks. The base RBC package is multi-mission ready, so it would just be a matter of porting over and updating the individual mission modules and identifying new modules for the LNext fleet.

There are expected to be at least several new inputs to RBC for LNext, including the downlinked Flight ATS, the MSD telemetry table, a new format for the contact schedule, etc. RBC will be required to constraint check the Ground and Flight ATS's against each other, which will be a new responsibility.

2.5.9 Mission Planning to DPAS

The CPAW to DPAS interface has very few mission-unique elements and could be readily used for one or multiple new observatories. Depending on the mission architecture changes however, some product formats may need to be adjusted (e.g., if the WRS-2 Grid were no longer used)

2.5.10 Calibration Opportunity Scheduling Tool

The Calibration Opportunity Scheduling Tool (COST) is an FOT-developed tool for automatically scheduling calibrations not scheduled by CPAW. The reuse percent of COST will be largely driven by how different the Landsat Instrument Suite (LandIS) calibration types are from L8 and L9. COST has been developed to be modular, allowing for rapid integration of new calibration types.

2.5.11 Attitude Planning

CPAW contains mission-agnostic software for planning and modelling observatory slews. However, the Landsat 8 / 9 observatories have several pre-defined ACS Modes (e.g., Solar / Lunar, Side Slither, Off-NADIR) which have been modeled in CPAW. If new mission(s) were to have a similar set of pre-defined ACS modes, they too would need to be added to the CPAW Attitude Planning software.

2.5.12 Automation

In the current LMOC, the MPS is the software system which requires the most manual interaction. CPAW application, and to a lesser degree STK Scheduler, still requires user interactions to navigate the GUI.

Additional resources could be used to update the product line software for CPAW to improve the ability to perform the nominal load building tasks via Application Programming Interface (API) calls. This would further the FOT's ability to automate all the tasks within the application itself.

Figure 18 below gives a summary of the manual edits that MPS has had to make during the load building process for L8 and L9 since 3/13/2023. Prior to that date, these metrics were not maintained consistently. There were 111 manual edits made during this time span over an estimated 41 load builds for each spacecraft, averaging to 1.4 edits per load build. 39 of these edits are related to issues which are currently resolved. 14 of the edits are related to special activities and ad-hoc or late-breaking updates, all of which are largely unavoidable and just part of the job for MPS. The remaining are ticketed and prioritized appropriately for closure, either as a work ticket in LMOCOPS project or as a software discrepancy in LMOCDEV project or both.

Of the edits listed in Figure 18, some can be grouped at a higher level. 36 of the edits are related to housekeeping management activities (L8 HKDB or L9 SSOH playbacks and recordings). This is not a native role for CPAW and there has been a learning curve for the development of this function within the software's Landsat plugin. This represents a pressure point for the CPAW that has to be resolved both for L8 and L9 operations and for the augmentation of the LMOC to support LNext.

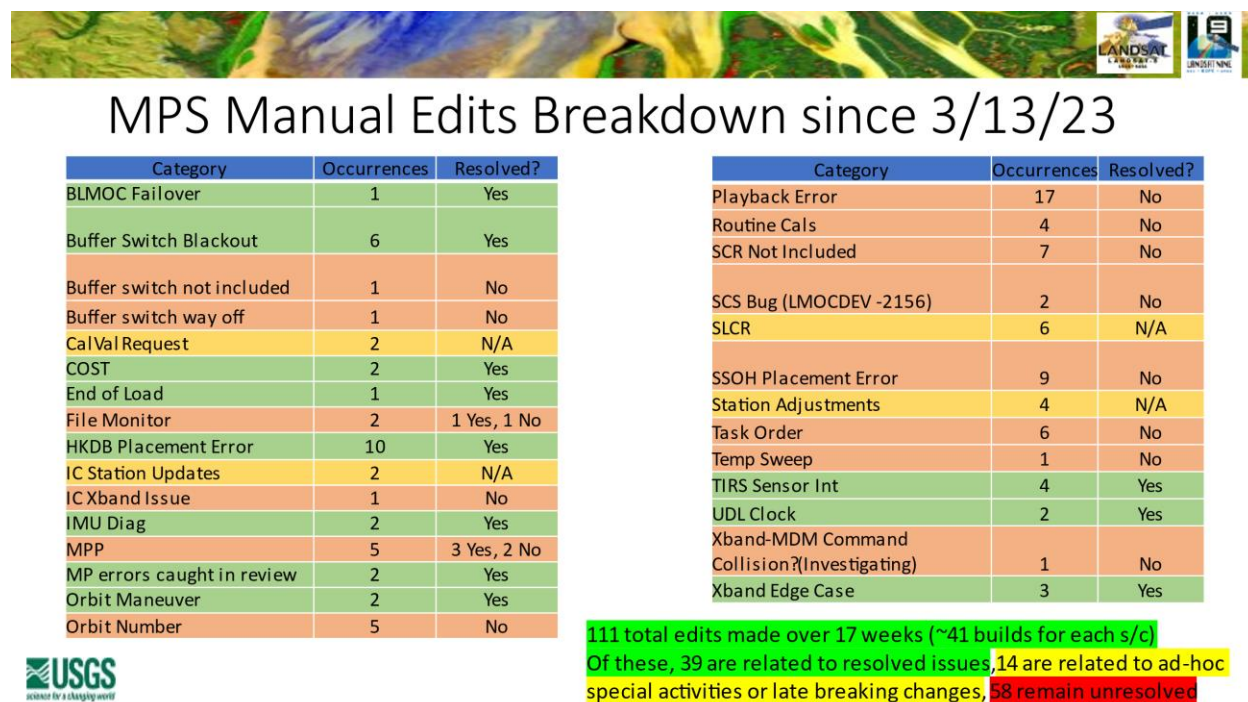


Figure 18: MPS Manual Edits made since 3/13/23

Eight of the edits are related to issues with the buffer switch. This is another non-native function of CPAW, but most of these issues have been resolved via software updates or MAL postprocessing. Four of the edits relate to routine calibrations, and 11 relate to non-routine calibrations and other special activities.

Looking ahead to a much more simple set of MPS responsibilities, there is a potential to enhance the automation of the process either with CPAW or with a replacement software.

2.6 Flight Dynamics Subsystem Scalability

The Flight Dynamics System (FDS) provides attitude estimation and sensor calibration, orbit prediction, maneuver planning, definitive attitude and orbit ephemeris generation, mission planning products, conjunction assessment support, and ground station acquisition data. The FDS system manages all Landsat observatories operated by the LMOC.

The following provides a high-level overview of the key functionality provided by FreeFlyer / MTASS / Meridian, its potential reuse for additional missions, and some particulars for where mission specific functionality has been configured or developed.

2.6.1 Orbit & Attitude Determination

Orbit and Attitude determination does not contain any mission specific code, however depending on the data formats available from spacecraft housekeeping, some pre-processing of input data may be required for FDS to process and generate the definitive products.

2.6.2 Orbit Control and Prediction

There is not expected to be any substantial software changes in these scripts, however if the new mission(s) were to have a non-chemical propulsion system, then some re-work of mission scripts may be required to model the new types. For example, using electric propulsion for orbit maintenance would likely be a net benefit to the mission, primarily by eliminating propellant as a life limiting item, but requires more significant changes to the FDS implementation.

Implementing a constellation of satellites with chemical thrusters would likely drive a higher level of automation to control the FOT level of effort.

2.6.3 Conjunction Assessment

The FDS conjunction assessment and mitigation services contain no mission-unique elements and could be readily used for the LNext observatories without modification.

The FOT has developed a suite of tools to support Expedited Risk Mitigation Maneuvers. These allow the team to execute an orbit maneuver within 12 hours of a new conjunction screening using the nominal process, and even quicker as the situation required. As the operations grows to support a constellation of five, this process will need to become more automated. There is nothing technical that prevents it from being automated today, but it would require an acceptance of a different risk posture (burning in the blind, without the ability to call off last second).

2.6.4 Attitude Prediction

Similar to CPAW, the FDS Attitude prediction software is mission-agnostic, and will only require script updates to model mission-defined ACS modes. All other changes would be made in software settings.

2.6.5 Attitude Control

Attitude Control generates the Attitude Window Table (AWT) which is a flight load file for performing Lunar Scans. Attitude Control also performs independent attitude constraint checking on activity plans. These load files are LEOStar-3 specific formats and would need re-developed if similar capability was required on the new observatories. Any new ACS modes would need to be defined in the FDS Attitude Control scripts so they could be modeled accurately.

2.6.6 Attitude Sensor Alignment and Calibration

Any new sensor types would need to be added to the scripts and new alignment quaternions would need to be configured in the software. Otherwise, there is little to no change required.

2.6.7 Product Generation

The FDS product generation services contain no mission-unique elements and could be readily used for the LNext observatories without modification. When generating daily products, each mission is run sequentially. Depending on the number of observatories added, two FDS systems running in parallel may be needed to meet daily product deadlines. Alternatively, as indicated above with CPAW load building, if mission loads are uplinked less frequently, this can be accommodated by staggering product generation days for the various missions.

2.6.8 Propellant Bookkeeping

Some small changes may be required depending on the propellant data available from the observatory.

2.6.9 Fleet Data Management

The FDS fleet data management services contain no mission-unique elements and could be readily used for the LNext observatories without modification.

2.6.10 Orbit and Attitude Validation

The FDS Orbit and Attitude Validation services are done using a real-time contact using the telemetry stream from the observatory. The data formatting and units would need to be configured in the system but otherwise the validation against predicted values is all mission agnostic.

2.6.11 Station Visibilities

The current FDS system generates the station visibilities for Landsat 8 and 9 by default but they are not used since this is a responsibility of STK scheduler. On Landsat 8 these visibilities were used by the Flexplan system successfully. For LNext since we are recommending a simplified MPS system which would no longer perform orbit prediction and we would once again utilize this capability in FDS for the LNext station visibilities. This is an out of the box capability of FreeFlyer and would not need any modification to support LNext other than identifying the ground sites and station masks.

2.7 LMOC Hardware Scalability

Functionality	Reuse	Particulars
Virtual Cluster	●	Mission Agnostic, Highly Scalable
Storage	●	Mission Agnostic, Highly Scalable
Networking	●	Mission Agnostic, Highly Scalable
User Interfaces	●	Mission Agnostic, Highly Scalable
Management	●	Mission Agnostic, Highly Scalable

Table 4: LMOC Hardware Scaling and Reuse by Function

2.7.1 Virtual Cluster

The LMOC is a Virtual Cluster operating approximately 145 virtual machines to support Landsat 8 / 9. While the infrastructure could be scaled up by adding additional hypervisor servers and joining them to the cluster we recommend instead that LNext capabilities are migrated to the cloud. For the small subset of subsystems that are required to be onsite (The Realtime Operations Consoles) there is more than adequate cluster resources to accommodate them. On an average day, the LMOC is only utilizing around 15% of its CPU resources and 40% of its RAM.

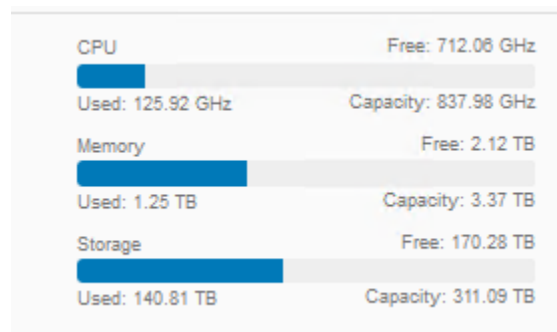


Figure 19: LMOC Virtual Cluster Resource Utilization

2.7.2 Storage

As discussed earlier, the LMOC storage will become a limiting factor when adding new missions to the system. The LMOC Data store is scalable, however scaling it up for a constellation would require an additional rack for disk storage and the associated networking equipment which is not desirable given the limited space in B32 C101C. Ideally all additional storage needs would be met by cloud services which would not require any additional on-prem hardware and the on-prem storage footprint could be reduced by possibly having a hybrid trending system that was both on-prem and in the cloud.

2.7.3 Networking

The LMOC network is built upon a 40Gbps backbone with 10Gbps external interface capabilities. The network has plenty of margin to scale up to support a new mission or a constellation without modification. This network is also sufficient to handle a hybrid LMOC/Cloud environment since most LMOC to Cloud interfaces would be file based and would not require significant bandwidth. Migrating the majority of the LNext capabilities into the cloud would also help alleviate the already strained 1Gbps connections we have to GSFC and EROS.

2.7.4 User Interface

The LMOC User Interface contains no mission specific functionality and can be used to support a single observatory or constellation of observatories. The interface could also be scaled readily by adding more zero clients and associated software licenses. There is no known max number of users or clients.

2.7.5 Management

The LMOC Infrastructure Management Software contains no mission specific functionality and can be used to support LNext with little to no modification.

2.8 LMOC Software Scalability

Functionality	Reuse Particulars	
Network Management		Mission Agnostic, Highly Scalable
Alerting		Mission Agnostic, Highly Scalable
Broker / Client Management		Mission Agnostic, Highly Scalable

Table 5: LMOC Software Scaling and Reuse by Function

2.8.1 COTS Software

The LMOC Infrastructure management tools do not contain any mission unique functionality and can be used without modification for additional spacecraft.

2.9 Analytical Modeler

Analytical modeling is done with the Joint Activity Simulation for Observatory and Network (JASON) model – a GOTS tool. The JASON model is an operational tool developed for LDCM and updated for the L9 mission. More details on the tool can be found in the Landsat 9 OpsCon Document and the Landsat 9 MOC OpsCon.

This system within LMOC is currently *inoperable*. Assessment: **REPLACE**

- JASON is hard coded to WRS-2 and not extensible or scalable to other missions. As a result, LMOC doesn't have a modeling capability for science acquisition management and impact assessments.
- While CPAW has an SSR model, it's not a mission projection model. Therefore, human analyses is what produces SSR statistics and trends for performance use.

- This approach is unable to perform any predictive modeling assessments as it's entirely predicated upon telemetry, ground stations, and ack and nak information.

New development will be needed for analytical modeling on the LNext project. A digital model of the data volume chain within LMOC would be substantially beneficial. This could also be accomplished as an applied version of the LNext model and simulation tool developed for the DPAS that can be extended to the MOC. For LNext, three different models or modes based on actual configuration of each vehicle may be necessary.

2.10 Configuration Management

Landsat 8 and 9 operations use Bit Bucket, Confluence, and JIRA as the CM tools for mission operations configured items, trouble tickets, and engineering file management (daily work and collaboration). Bitbucket is the versioning tool. Confluence and JIRA are used for product and other documentation and trouble ticket tracking.

USGS also utilizes Docushare and, recently, Microsoft collaboration tools (Teams and Sharepoint, for example). Docushare contains mission and program level documents, from management plans to policy documents. It also contains the historical data archive from Landsat 8 and 9 NASA development – all documents handed over during transition from NASA to USGS for long term operations. Sharepoint is primarily used for collaboration at USGS.

HW and SW Configured Items are tracked via various tools, which include excel spreadsheets and Bitbucket. This is the traditional configuration of software, hardware, products and documents related to the ground system and operations. Mission Operations Configured Items (MOCIs) are also tracked via Bit Bucket, Confluence, and Jira. As currently implemented (as of 2023), it has some limitations:

- It is not a snapshot of the system (e.g., reference image)
- It does not contain an easy to use or find Observatory configuration history (E.g. changes to all onboard table parameters) – no summary or situational awareness tools purposefully built for the FOT to use quickly
 - Capability has been added over time to fill gaps (eg. ConfigCheck), but is spread across the system and in config files and version management tools.
- No built in functionality to confirm ground models match observatory models, as needed
 - For example, earth models were different between spacecraft and FDS.

As a result of these limitations, work has been done across the project to fill in the gaps, and create new policy documents for all current and future missions to ensure consistent application of configuration management practices.

LNext needs to understand differences between each satellite in constellation management as it pertains to traceability of features and changes for all ops activities. LNext also needs core capability to monitor changes in the system, and have a complete history of the changes. Therefore, there is a need to expand the traditional CM definition from tracking HW, SW, and MOCI, to include state checking, memory management and tracking, and system to system (flight to flight to sim to ground, etc.) difference monitoring and tracking.